# Lichao Shen

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## INTERESTS

Human-Computer Interaction

Human Augmentation, Virtual Reality, Telepresence, Physical & Embodied Interface, Interaction Design

## EDUCATION

**Master.** Media Design

2015 – 2018 Keio University, tokyo, japan

- Human-Computer Interaction, Graduate School of Media Design.
- GPA 3.84 / 4.33. English instruction.
- Global Innovation Design: exchange study at Pratt Institute, US, Royal College of Art and Imperial College London, UK.
- Advisor: Prof. Kouta Minamizawa, Co-advisor: Prof. Kai Kunze.
- Thesis: *Limitless Oculus: Visual Expansion by Animal-Inspired Modification to Visuomotor Mechanism*.

**Bachelor.** Engineering

2010 – 2014 Beihang University, beijing, china

- Industrial Design, School of Mechanical Engineering and Automation.
- GPA 3.6 / 4.0. Outstanding graduate.

# RESEARCH EXPERIENCE

### DKU HCI Lab.

2023 – 2024 Duke Kunshan University, kunshan, china

- Director: Prof. Xin Tong. Collaborative research.
- Led AIGC for ASD: personalized, adaptive, and interactive storytelling for autistic children.
- Designed and prototyped an intelligent assistive system with multimodal interfaces and contextualized stories powered by real-time generative AI, to assist personalization and educational interventions focused on social-emotional learning, considering multi-stakeholder and accessibility.

### Shared Reality Lab.

2021 – 2023 McGill University, montréal, canada

- Director: Prof. Jeremy Cooperstock. Doctoral program, incomplete.
- Led Affective Telepresence: multimodal recognition and representations for sharing affect remotely.
- Developed an intelligent framework for affect detection, representation, and communication by mapping physiological, bodily, and environmental signals to multisensory feedback, and robotic, physical and embodied expressions, to recreate the sense of shared ambiance. Investigated non-intrusive sensing, nonverbal, passive, and implicit interaction, and human-robot interaction during shared activities.

### Privacy and Security Lab.

2022 University of Maryland, baltimore county, us

- Director: Prof. Yaxing Yao. Collaborative research.
- Led VR Privacy: explore the design space of privacy notifications in virtual environments.
- Proposed a taxonomy to design notifications and notices indicating privacy status in immersive virtual reality; Designed and prototyped forms and methods, considering privacy policy and design theory.

### Accessible and Immersive Realities Lab.

2020 Rochester Institute of Technology, rochester, us

- Director: Prof. Roshan Peiris. Collaborative research.
- Led Visuomotor Effect: visual expansion and sensorimotor remapping through wearable robotics.
- Developed wearable robotic neck substitution system with augmented reality interface; Design and conducted experiments to evaluate and quantify the influence of modifications to natural visuomotor coordination, involving visual and spatial perception, proprioception, and hand-eye coordination.

### Empathic Computing Lab.

2019 University of Auckland, auckland, new zealand

- Director: Prof. Mark Billinghurst. Visiting research.
- Led Virtual Lab: collaborative virtual workspace for real-time sharing surroundings and activities.
- Developed mobile platform for remote access and operation in virtual workspace; Designed the camera array and sensor cluster integrating depth vision, and eye hand tracking, to support scene reconstruction, 3D panorama, real-time visualization, and recognition for remote collaboration in mixed reality.

### KMD Research Institute.

2018 Keio University, yokohama, japan

- Director: Prof. Kouta Minamizawa. Post-graduation research.
- Led Metamorphosis: immersive embodiment into nonhuman robot by sensorimotor reconstruction.
- Developed interactive prototypes for sensory and motor alteration, and human augmentation focused on nonhuman body schema, through biomimetic approaches, combining extended reality, robotics, and wearables; Implemented haptic and thermal feedback, embodied and tangible interaction, and interfaces of brain, skin, and muscle, to extend the boundary of Telexistence and Embodied Media.

# INDUSTRY EXPERIENCE

**Co-founder, Technical Product Lead.**

2025 – 2026 Robo-Data Vehicle, beijing, china

- Led startup strategy and product development of AI data infrastructure dedicate to Embodied AI, Physical AI, VLA, world model, and robotics.
- Led early-stage research and development. Designed core data pipeline and platform: collect, process, train, generate, and trade; Implemented prototype demonstration, software development, and the integration of computer vision algorithms and language models.

**Product Manager.**

2024 Zhejiang University Nanchang Institute, nanchang, china

- Led the product design and development of wearable intelligence, AI agent system on XR smart glasses;
- Designed and defined product requirements, core features, user experience, and the architecture of LLM-based AI agent operating system and application ecosystem; Coordinated developer teams to deliver prototypes, customer deployments, and optimization compatibility to multiply hardware.

**Product and Solution Manager.**

2019 – 2021 China Mobile Shanghai Research, shanghai, china

- Led the full product life-cycle of enterprise AR solutions and in-vehicle electronic products, from market research and business analysis to product design and development, product management and operation, and technical support.
- Designed and developed core platforms products, defining application scenario, product features, hardware and software specifications; Coordinated developer and engineer teams and OEM partners.

# PUBLICATIONS

**Conference Publications**

2020 **Lichao Shen**, MHD Yamen Saraiji, Kai Kunze, Kouta Minamizawa, and Roshan Lalintha Peiris. 2020. Visuomotor Influence of Attached Robotic Neck Augmentation. In *Symposium on Spatial User Interaction* (SUI '20). ACM, New York, NY, USA, Article 14, 1–10. DOI:https://doi.org/10.1145/3385959.3418460

2018 MHD Yamen Saraiji, Roshan Lalintha Peiris, **Lichao Shen**, Kouta Minamizawa, and Susumu Tachi. 2018. Ambient: Facial Thermal Feedback in Remotely Operated Applications. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (CHI EA '18). ACM, New York, NY, USA, Paper D321, 1–4. DOI:https://doi.org/10.1145/3170427.3186483

2018 **Lichao Shen**, MHD Yamen Saraiji, Kai Kunze, and Kouta Minamizawa. 2018. *Unconstrained Neck: Omnidirectional Observation from an Extra Robotic Neck*. In Proceedings of the 9th Augmented Human International Conference (AH '18). ACM, New York, NY, USA, Article 38, 1–2. DOI:https://doi.org/10.1145/3174910.3174955
**[Best Demo Award]**

**Patents**

2020 CN211319386U. Vehicle-road cooperative system

2021 CN214507271U. Projection terminal.

# SERVICES

**Peer Review**

- CHI 2026, IMX 2026
- IDC 2024, UIST 2024, CSCW 2024, SAIS 2024, ISWC 2024
- WHC 2023, ISMAR 2023
- IDC 2022, EICS 2022

# EXHIBITIONS

2014 Auto China, 13th Beijing International Automobile Exhibition, *AERO* Racing Car

2013 Formula Student China, 8th of 53 teams, *AERO* Racing Car

2013 *City, Museum, Tokyo* Design Show, *Hahaki* Postcard System

# HONORS

**Awards**

2018 Best Demo Award, Augmented Human 9th International Conference

2013 Excellent Prize, *Gifts of Fuyang* Design Competition

2013 Finalist , Red Dot Design Award

2011 Second Prize, *Fengru Cup* Student Innovation Competition

2011 Excellent Prize, *Badaling Cup* Souvenir of Great Wall Design Competition

**Titles**

2014 Outstanding Graduate

2013 Outstanding Student Leader

2011 Merit Student

**Scholarships**

2021 McGill Engineering Doctoral Award

2013 Scholarship of Academic Performance (3 years)

2013 Scholarship of Social Duty

2011 Scholarship of Science and Technology Competition

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